Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 04-Apr-2026

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1. Administrative & Study Identification

Full Study Title: An Open-Label, Parallel-Group Study to Evaluate Safety, Tolerability, Pharmacokinetics, and Pharmacodynamic Effects of Ocrelizumab in Children and Adolescents With Relapsing-Remitting Multiple Sclerosis **Study Title:** A Study of Ocrelizumab in Children and Adolescents With Relapsing-Remitting Multiple Sclerosis **Short Title/Acronym:** OPERETTA I **Protocol Number:** WA39085 (2016-002667-34) **NCT Number:** NCT04075266 **Posting ID:** 207230974578718374 **Trial Status:** Active, Not Recruiting **Sponsor/Company Name:** F. Hoffmann-La Roche Ltd / Genentech / Roche **Investigational Product:** Ocrelizumab (Ocrevus), Anti-CD20 monoclonal antibody **Phase of Development:** Phase 2 **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, pharmacokinetics (PK), and pharmacodynamic (PD) properties of ocrelizumab to determine the appropriate dose for children and adolescents with relapsing-remitting multiple sclerosis (RRMS). **Secondary Objectives:** To evaluate the number of new/enlarging T2 lesions on MRI, annualized relapse rate (ARR), disability progression, and long-term efficacy and safety over an optional extension period.

2.2 Background & Rationale To address the critical need for effective disease-modifying therapies (DMTs) in pediatric-onset multiple sclerosis (POMS). Multiple Sclerosis is a progressive autoimmune disorder affecting the central nervous system resulting in demyelination. Patients develop physical and cognitive impairments that correspond with the affected nerve fibers. Children and adolescents tend to have a more highly inflammatory disease course, higher relapse rates, and more aggressive MRI activity compared to adult-onset MS. This study establishes the appropriate dosing regimen for younger patients prior to proceeding to a definitive Phase III efficacy trial.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Parallel-group (two dose cohorts based on body weight) **Blinding:** Open-label **Randomization:** N/A (Assigned to cohorts based on body weight cutoff)

3.2 Planned Enrollment & Duration Target Enrollment: 23 enrolled pediatric participants **Number of Sites:** Multicenter across 3 Countries **Estimated Study Start:** 2019 **Estimated Primary Completion:** October 2023 (Primary analysis cutoff)

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 10 to ≤ 18 years. 2. Diagnosis of relapsing-remitting multiple sclerosis (RRMS). 3. Body weight ≥ 25 kg. 4. Expanded Disability Status Scale (EDSS) at screening: 0-5.5, inclusive. 5. Children and adolescents must have received all childhood required vaccinations. 6. Female participants of childbearing potential must agree to either remain completely abstinent or to use reliable means of contraception. 7. Neurologic stability for ≥ 30 days prior to screening, and between screening and baseline. 8. Participants naive to prior disease-modifying therapy (DMT); OR participants who have had at least 6 contiguous months of DMT within the past 1 year must have evidence of disease activity occurring after the full 6-month course of treatment (at least one relapse or ≥ 1 Gd-enhancing lesion(s) on a T1-weighted brain MRI).

4.2 Exclusion Criteria 1. Known presence or suspicion of other neurologic disorders that may mimic MS, including, but not limited to, acute disseminated encephalomyelitis (ADEM), neuromyelitis optica or neuromyelitis optica spectrum disorders, and any neurologic, somatic, or metabolic condition that could interfere with brain function or normal cognitive or neurological development. 2. Patients that are aquaporin 4 positive and myelin oligodendrocyte glycoprotein (MOG) antibody positive. 3. In case of an ADEM-like appearance of the first MS attack, a second attack with clear MS-like features is required. 4. Infection requiring hospitalization or treatment with IV anti-infective agents. 5. History or known presence of recurrent or chronic infection (e.g., HIV, syphilis, tuberculosis). 6. Receipt of a live or live-attenuated vaccine within 6 weeks prior to treatment allocation. 7. History or laboratory evidence of coagulation disorders. 8. Peripheral venous access that precludes IV administration and venous blood sampling. 9. Inability to complete a magnetic resonance imaging (MRI) scan. 10. History of cancer, including solid tumors, hematologic malignancies, and carcinoma in situ. 11. History of a severe allergic or anaphylactic reaction to humanized or murine monoclonal antibody (mAbs) or known hypersensitivity to any component of ocrelizumab solution. 12. Previous treatment with B-cell-targeted therapies. 13. Percentage of CD4 < 30%. 14. Absolute Neutrophil Count < 1.5x1000/microliter. 15. Lymphocyte count below the lower limit of normal (LLN) for age- and sex-specific reference range.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Group 1 (weight < 40 kg): 300 mg of ocrelizumab; Group 2 (weight $\ge$ 40 kg): 600 mg of ocrelizumab. First dose is given as two separate infusions at least 2 weeks apart. Subsequent doses are administered as one single infusion every 24 weeks. *(Note: Data analysis resulted in shifting the threshold to 35 kg for future trials)* **Route of Administration:** Intravenous (IV) Infusion **Duration of Treatment:** 24-week dose-exploration period followed by an optional extension period of 264 weeks (5.5 years). **ATC Code:** L04AA36

5.2 Concomitant & Prohibited Therapy Permitted: Mandatory prophylactic medications prior to infusion (e.g., antihistamines like diphenhydramine, and weight-adjusted methylprednisolone) to mitigate infusion-associated reactions. **Prohibited:** Concurrent administration of other investigational MS disease-modifying therapies (DMTs).

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to 8 weeks	Informed Consent, Medical History, MRI, EDSS, Labs
Baseline	Day 0	Cohort Assignment, First Infusion, Pre-medications, PK/PD draw
Interim	Q4W to Q12W	Every 4 weeks in the first 6 months; every 12 weeks thereafter. Safety Labs, EDSS Assessment, PK/PD tracking
End of Study	Year 5.5	Final MRI, Clinical assessment, safety follow-up for a minimum of 48 weeks post-last dose

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Dose Exploration Period: Area Under the Concentration Versus Time Curve of Ocrelizumab:** Assessed using a population PK model to simulate concentration-time course and predict individual AUC, analyzed per body weight range. * **Dose Exploration Period: Maximum Concentration (Cmax) of Ocrelizumab:** Assessed using a population PK model to simulate concentration-time course and predict individual Cmax, analyzed per body weight range. * **Dose Exploration Period: Levels of CD 19+ B-cell Count in Blood.**

7.2 Secondary & Exploratory Endpoints * Number of Participants With Adverse Events (AEs). * Dose Exploration Period: Level of Circulating T Cells and Natural Killer (NK) Cells. * Optional Open-Label Extension (OOE) Period: Level of Circulating T Cells and NK Cells. * Annualized Relapse Rate (ARR) and changes in EDSS score. * Proportion of patients free of new/enlarging T2 lesions on brain MRI.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring during clinical visits every 4-12 weeks, specifically for infusion-related reactions, AEs, and susceptibility to mild-to-moderate infections. **Serious Adverse Events (SAEs):** Reported standardly within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Independent internal and external safety monitoring committees providing continuous clinical oversight.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety/PK/PD Sets (All patients who received at least one dose of study medication). **Sample Size Justification:** A sample of ~23 pediatric subjects is sufficient for exploratory PK/PD characterization to determine non-inferior exposure compared to adult Phase III trials. **Handling of Missing Data:** Last Observation Carried Forward (LOCF) or Mixed-Effect Model Repeated Measure (MMRM) for continuous secondary endpoints.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Comprehensive site monitoring including continuous verification of informed consent/assent and source data verification for the pediatric population. **Audit Trail:** Validated eCRF systems, stringent laboratory tracking for PK/PD samples, and data clarification mechanisms. **Regulatory Documentation:** EMA Product Information

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11. Study Identifiers & Governance

Primary ID: WA39085 **Registry Number:** NCT04075266 **Posting ID:** 207230974578718374 **Sponsor/Lead Organization:** Hoffmann-La Roche / Roche **Study Title:** A Study of Ocrelizumab in Children and Adolescents With Relapsing-Remitting Multiple Sclerosis **Official Regulatory Title:** An Open-Label, Parallel-Group Study to Evaluate Safety, Tolerability, Pharmacokinetics, and Pharmacodynamic Effects of Ocrelizumab in Children and Adolescents With Relapsing-Remitting Multiple Sclerosis **Status:** Active, Not Recruiting

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12. Clinical Framework (Multiple Sclerosis Specific)

Disease Area: Multiple Sclerosis **Primary Condition:** Relapsing-Remitting Multiple Sclerosis (RRMS) **Semantic Type:** Disease or Syndrome **Diagnosis Threshold:** Clinically definite POMS / RRMS, EDSS 0-5.5 **Medical Methodology:** B-Cell Targeting Therapy (Anti-CD20 Monoclonal Antibody) **Optimization Target:** Rapid and sustained CD19+ B-cell depletion / No Evidence of Disease Activity (NEDA) **Terminology Codes:** * **MeSH Code:** D009103 * **SNOMED ID:** 24700007 * **ATC Code:** L04AA36

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13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for RRMS disease-modifying therapy **Education Protocol (HCP):** Site-specific training on pediatric MS management and infusion protocol safety. **Education Protocol (Patient):** Age-appropriate assent materials, infusion-reaction awareness, and self-care training. **Quality Audit Mechanism:** Continuous AE/SAE reporting, EDSS rater certification tracking, and centralized MRI reading.

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14. Observation & Follow-Up Schedule

Total Duration: 24 weeks primary + 264 weeks extension (5.5 years) + 48 weeks safety follow-up **Onsite Visit Cadence:** Every 4 weeks for the first 6 months, then every 12 weeks **Remote Monitoring Frequency:** Supplemental tele-visits as clinically indicated **Checklist Review Interval:** Continuous safety monitoring aligned with every 12-week infusion cadence

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15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | | Clinical Operations Lead | [Name] |** ___ **|** ***[DD-MMM-YYYY]* | |** ***Lead Statistician* | *[Name]* |** _____ | [DD-MMM-YYYY] |